PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.



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Mechanical
3D CAD modeling, mechanism design, and fabrication.

Electronics
Schematic design, circuit wiring, and hardware assembly.

Embedded Control
Microcontrollers, sensors, and actuator integration.

Prototyping
Physical build, functional testing, and validation workflows.

SCAN KODE QR ATAU KLIK DI MANA SAJA UNTUK BUKA PORTOFOLIO WEB

https://portobayu.vercel.app/

Portofolio Interaktif: Video Pengujian Lengan Robot & KRAI, Skematik & Model 3D CAD

Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom

01 / FINAL PROJECT



Robot Manipulator 5-DOF

5-DOF articulated robotic arm prototype powered by an STM32F411 microcontroller and 3D-printed cycloidal gearboxes. Integrates AS5600 magnetic encoders in a closed loop to reduce backlash, achieving 79.8% accuracy across 250–1,000 g loads.

STM32F411 MCU	NEMA 17 Stepper
TB6600 Driver	AS5600 Sensor
TCA9548A Mux	Cycloidal Gearbox

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02 / CAPSTONE DESIGN



Pendingin Panel Surya Otomatis

Automated water spray cooling system to lower solar panel surface temperature driven by Arduino Uno. Uses temperature sensing and relay control to trigger a submersible pump above >30°C and shut down below <25°C to restore efficiency.

Arduino Uno R3	RTD Temp Sensor
5V Relay Module	12V Water Pump
16x2 I2C LCD	12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



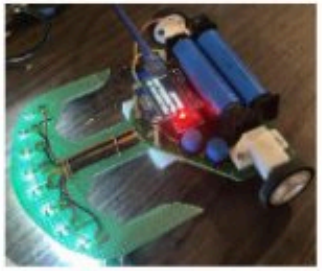
Kompetisi Robot Basket

Mobile basketball robot for Kontes Robot ABU Indonesia (KRAI). Served as a core mechanical team member designing the shooter, dribbler, and drive wheel configuration, advancing to the National Stage at ITB Jatinangor.

6-Bar Pneumatics	24V Solenoid
RS775 DC Motor	Omniwheel Drive
Limit Switch & IR	Air Regulator

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04 / PROYEK MANDIRI



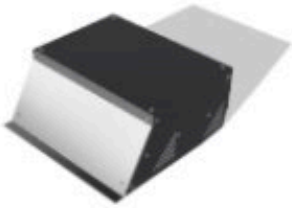
Robot Line Follower

Autonomous line-tracking robot featuring differential drive steering. Equipped with an 8-sensor bar built from LED and photodiode pairs, an Arduino Nano microcontroller, a TB6612FNG driver, and micro N20 metal gear motors.

Photodiode Array	Arduino Nano
TB6612 Driver	N20 Gear Motor
0.96" I2C OLED	2S Li-ion Battery

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05 / PROYEK MANDIRI



Robot Sumo

1 kg competition-class sumo robot (20 × 20 cm) built on an acrylic chassis with a front aluminum scoop. Delivers 4 kg push force via 4 PG24 planetary gear motors, steered wirelessly from a smartphone via Bluetooth connection.

Arduino Uno	4× PG24 Motors
L298N Driver	HC-05 Bluetooth
3S LiPo Battery	Acrylic Chassis

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KLIK SALAH SATU KARTU PROYEK DI ATAS ATAU SCAN KODE QR

https://portobayu.vercel.app/ · Dokumentasi Teknis, Video Validasi, Metrik & Model 3D

01. Robot Manipulator 5-DOF | 02. Pendingin Panel Surya | 03. Robot Basket KRAI | 04. Line Follower | 05. Robot Sumo

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